

Executive Summary

Hayes Diagnostics is developing a next-generation diagnostic platform designed to dramatically reduce the time, cost, and complexity of identifying bacterial pathogens. The company addresses a critical failure in today's \$24B+ global infection diagnostics market, where slow, labor-intensive, and error-prone microbiologic methods lead to delayed treatment, increased morbidity, prolonged treatment, rising healthcare costs, and increasing antibiotic resistance.

The Problem

Current diagnostic testing for bacterial infections is fundamentally broken. Gold-standard methods such as microbial culture can take up to two weeks to deliver results and fail to identify the causative pathogen up to one-third of implant related infections. During this delay, clinicians are forced to make treatment decisions with incomplete information, often prescribing broad-spectrum antibiotics that prolong illness, increase adverse events, morbidity and hospital stays, drive up costs, and contribute to antibiotic resistance.

The process itself is expensive and fragile, relying on a long chain of specialized personnel, large laboratory infrastructure, and time-sensitive sample handling—each step introducing cost, delay, and risk of error. Despite decades of incremental improvement, the industry has adapted to these medically unacceptable limitations simply because a faster, accurate alternative has not existed.

The Hayes Diagnostics Solution

Hayes Diagnostics' patented technology is a rapid, highly accurate diagnostic platform, which can identify targeted pathogens using their intrinsic physical properties. This streamlines pathogen detection while minimizing human intervention and laboratory complexity. The technology is designed to deliver actionable diagnostic results significantly faster than traditional culture-based methods, while achieving high sensitivity and specificity. By reducing dependence on centralized labs and complex workflows, Hayes Diagnostics enables earlier, targeted treatment decisions, improving patient outcomes while lowering total system cost.

Competitive Landscape

Existing diagnostic solutions fall into three main categories:

- **Microbial culture:** highly trusted but slow, labor-intensive, and prone to failure.
- **Molecular and immunoassay tests (PCR, ELISA):** faster but expensive, infrastructure-heavy, and prone to false positives.
- **Rapid strip or point-of-care tests:** inexpensive and fast but have long, expensive development.

Hayes Diagnostics bridges the gap by combining speed, accuracy, and operational simplicity into a single platform—addressing the core limitations of each existing category rather than trading one weakness for another.

Customer Value Proposition

- **Clinicians** gain faster, more reliable diagnostic answers, enabling precise treatment decisions and reducing unguided antibiotic use.
- **Hospitals and healthcare systems** benefit from lower lab costs, reduced length of stay, and improved operational efficiency.
- **Payers** see reduced downstream costs from complications, prolonged care, and ineffective treatments.
- **Patients** experience faster diagnosis, better outcomes, and fewer adverse effects.

Business Development Progress

Hayes Diagnostics is currently in the early stages of development, focused on technology validation, platform refinement, and stakeholder engagement. Initial concept development and market analysis have confirmed a large unmet need and strong demand for a rapid, accurate diagnostic solution. The company is actively advancing toward technology validation and preparing for strategic partnerships, pilot studies, and future regulatory approval.